

Empirical Software Engineering

Write your answers directly on these pages (there's always a risk that loose papers disappear)—use the back also if possible (if they've printed at the back then use other papers, but mark them clearly with your name). I'll be at the written exam twice (first time after approximately one hour).

On September 9 at 10.00 you are welcome to room 6217 in EDIT (Johanneberg) with questions about the grading. Before the meeting you *must* send Richard an email clearly pointing out where you think the error is, what you wrote, and why you believe the grading was not correct. If I don't receive such an email before 10.00 on September 8, then I will not meet with you.

Probability theory is nothing but common sense reduced to calculation.

— Pierre-Simon Laplace

Grade 3: 44 points; ~50%

Grade 4: 61 points; ~70%

Grade 5: 78 points; ~90%

Maximum: 87 points

Question 1 :

(8p) Over the years Bayesian data analysis has evolved and spread to all natural sciences. There are several reasons for this, e.g., a principled way to incorporate prior knowledge, increase in computational power making Markov chain Monte Carlo a viable option for sampling, and probabilistic programming languages gaining ground. However, not until lately have statisticians developed guidelines researchers can follow to systematically design Bayesian models.

In *your* opinion, what **key steps** are compulsory when conducting Bayesian data analysis? Please **explain** what steps one take when designing models, so that we ultimately can place *some* confidence in the results.

You can either draw a flowchart and explain each step, or write a numbered list explaining each step.

(It's ok to write on the backside, if they haven't printed on the backside again...)

Solution: There is no standard. However, quite recently Gelman *et al.* (arXiv:2011.01808) published a manuscript on arXiv explaining what they believed were the key parts in Bayesian data analysis. In short, a lot of the things they talk about in that paper are things we have covered in this course.

In order to get 8 points there need to be a minimum of 4 areas that are listed and explained, e.g., prior or posterior predictive checks, diagnostics, model comparison, etc.

Start with a null model, do prior checks, check diagnostics, do posterior checks, then conduct inferential statistics if needed. Also, it would be good if you mention **comparisons** of models and that it is an **iterative** approach.

Question 2 :

(4p) Underfitting and overfitting are two concepts not always stressed a lot with black-box machine learning approaches. In this course, however, you've probably heard me talk about these concepts a hundred times...

What happens when you underfit and overfit, i.e., **how does this bias the results?** What are some principled **ways to deal with** under- and overfitting?

Solution: One straightforward way to discuss this is to show two simple models, one where we only have a general intercept α (complete pooling), and one where you estimate an intercept for each cluster in the data (varying intercept), but where we don't share information between the clusters (think of the pond example in the coursebook), i.e., no pooling.

This question can be answered in many different ways.

$$y_i \sim \text{Binomial}(n, p_i)$$

$$\text{logit}(p_i) = \alpha$$

$$\alpha \sim \text{Normal}(0, 2.5)$$

above complete pooling, below no pooling

$$y_i \sim \text{Binomial}(n, p_i)$$

$$\text{logit}(p_i) = \alpha_{\text{CLUSTER}[j]}$$

$$\alpha_j \sim \text{Normal}(0, 2)$$

$$\text{for } j = 1, \dots, n$$

Question 3 :

(10p) To understand **how team size affects team maturity** the following data was collected:

Team	Team size	SPI	Team Maturity
1	5	67%	Level 2
2	15	33%	Level 1
3	11	49%	Level 1
4	7	90%	Level 2
$\vdots$	$\vdots$	$\vdots$	$\vdots$

The experiment started with assuming that planning effectiveness and team maturity has a very strong association. For planning effectiveness they used schedule performance indicator (SPI) as a latent variable, i.e., a variable representing planning effectiveness somehow (this is actually a common way to measure performance in industry):

$$\text{Schedule Performance Indicator} = (\text{Completed points} / \text{Planned points})$$

As you can see above, SPI is a ratio (0–100).

Based on the result, if SPI is more than 50% they are classified as a team with high team maturity (Level 2). If less than 50% they are classified as a team with low psychological safety (Level 1).

With the above data, the firm wants to use your knowledge to understand the association between team size and team maturity.

- Write down the mathematical model definition for this prediction using any variable names and priors of your choice.
- State the ontological and epistemological reasons for your likelihood. Remember to clearly state and justify the choices and assumptions regarding your model.

Solution: This one is actually a bit trickier than it looks like at first glance. Sure, we know that we should design a model where SPI **or** team maturity should be used as an outcome, but which one should we use?

SPI is a continuous number from 0 to 100, while team maturity can be seen as a dichotomous variable (basically 1/0) or a monotonically increasing variable. Hence, in the first case a Beta regression could be a possibility, while in the latter a logistic regression could be an option. The lesson here is that we should probably use model comparison to look at out of sample prediction capabilities.

I would be very hesitant in using team maturity as an outcome. Why? Well, we have taken a continuous value (SPI) and made it into a dichotomous value. In short, from an information theoretical point of view we are shooting ourselves in the foot. Never bin your variables! Even worse is that SPI in itself is also a latent variable...

To get max point on this question the answer **must** contain a reasoning that we should not use dichotomous variables as a replacement for continuous variables.

Question 4 :

(3p) What is **epistemological justification** and how does it differ from **ontological justification**, when we design models and choose likelihoods? Please provide **an example** where you **argue** epistemological and ontological reasons for a likelihood.

Solution: Epistemological, rooted in information theory and the concept of maximum entropy distributions. Ontological, the way nature works, i.e., a physical assumption about the world.

One example could be the **Normal** likelihood for real (continuous) numbers where we have additive noise in the process (ontological), and where we know that in such cases **Normal** is the maximum entropy distribution.

Question 5 :

(4p) When diagnosing Markov chains, we often look at several diagnostics to form an opinion of how well things have gone. Name four diagnostics we commonly use? What do we **look for** in each diagnostics (i.e, what thresholds or signs do we look for)? Finally, **what do they tell us**?

Solution:

Effective sample size: A measure of the efficiency of the chains. 10% of total sample size (after warmup) or at least 200 for each parameter.

$\hat{R}$: A measure of the between and the within variance of our chains. Indicates if we have reached a stationary posterior probability distribution. Should approach 1.0 and be below 1.01.

Traceplots: Check if the chains are mixing well. Are they converging around the same parameter space?

MCSE: Markov chain standard error or Standard Error for short.

E-BFMI: etc.

Divergent chains: ...

Question 6 :

(4p) **Explain** the four main benefits of using multilevel models.

Solution:

- Improved estimates for repeat sampling. When more than one observation arises from the same individual, location, or time, then traditional, single-level models either maximally underfit or overfit the data.
- Improved estimates for imbalance in sampling. When some individuals, locations, or times are sampled more than others, multilevel models automatically cope with differing uncertainty across these clusters. This prevents over-sampled clusters from unfairly dominating inference.
- Estimates of variation. If our research questions include variation among individuals or other groups within the data, then multilevel models are a big help, because they model variation explicitly.
- Avoid averaging, retain variation. Frequently, scholars pre-average some data to construct variables. This can be dangerous, because averaging removes variation, and there are also typically several different ways to perform the averaging. Averaging therefore both manufactures false confidence and introduces arbitrary data transformations. Multilevel models allow us to preserve the uncertainty and avoid data transformations.

Question 7 :

(4p) Below you see a Generalized Linear Model

$$y_i \sim \text{Binomial}(m, p_i)$$
$$f(p_i) = \alpha + \beta x_i$$

What is $f()$, and **how does it work**? Provide at least **two examples** of $f()$.

Solution: Generalized linear models need a *link function*, because rarely is there a “ μ ”, a parameter describing the average outcome, and rarely are parameters unbounded in both directions, like μ is. For example, the shape of the binomial distribution is determined, like the Gaussian, by two parameters. But unlike the Gaussian, neither of these parameters is the mean. Instead, the mean outcome is np , which is a function of both parameters.

Since n is usually known (but not always), it is most common to attach a linear model to the unknown part, p . But p is a probability mass, so p_i must lie between zero and one. But there’s nothing to stop the linear model $\alpha + \beta x_i$ from falling below zero or exceeding one.

The link function f provides a solution to this common problem. A link function’s job is to map the linear space of a model like $\alpha + \beta x_i$ onto the non-linear space of a parameter.

Question 8 :

(5p) Write an example mathematical model formula for a Poisson regression model with two different kinds of varying intercepts, also known as a cross-classified model.

Solution: See pp. 415 in course book for an example and don't forget to use the sub i notation!

SOLUTION

Question 9 :

(4p) **Explain** the terms in your own words:

- prior
- posterior
- information entropy
- instrumental variable

Solution: The first three are easiest to explain using the formulas (Bayes' theorem and information entropy). However, one could make a more high-level conceptual explanation.

The final one can be explained in different ways, e.g., in causal terms, an instrumental variable is a variable that acts like a natural experiment on the exposure.

Or in other words, it is a third variable, Z , used in regression analysis when you have endogenous variables—variables that are influenced by other variables in the model, i.e., you use it to account for unexpected behavior between variables. Using an instrumental variable to identify the hidden (unobserved) correlation allows you to see the true correlation between the explanatory variable X and response variable, Y .

Let's say you had two correlated variables that you wanted to regress: X and Y . Their correlation might be described by a third variable Z , which is associated with X in some way. Z is also associated with Y but only through Y 's direct association with X . For example, let's say you wanted to investigate the link between depression (X) and smoking (Y). Lack of job opportunities (Z) could lead to depression, but it is only associated with smoking through its association with depression (i.e. there isn't a direct correlation between lack of job opportunities and smoking). This third variable, Z (lack of job opportunities), can generally be used as an instrumental variable if it can be measured and its behavior can be accounted for.

Question 10 :

(4p) What are the **two kinds** of varying effects (random effects)? **Explain** the effect they have on a statistical model.

Also, can you use **continuous values** for varying effects? If yes, how?

Solution: Intercepts and slopes. Our models often make a jump in prediction capability when we add both. Together they are often called varying effects.

A varying intercept means we estimate an intercept for each, e.g., a subject in an experiment. Varying slopes allows us to also estimate a separate slope for each subject.

These type of models often result in many parameters that need to be estimated. That is however usually OK, as long as we use partial pooling and an information criteria to keep an eye on overfitting (remember: the more complex models will be penalized).

Yes, you can use continuous values in random effects, e.g., Gaussian Process.

Question 11 :

(8p) What are the **four elemental relations** on which any Directed Acyclic Graph can be explained?

Please draw the four relations. **Explain** what they mean (i.e., by explaining if one should condition or not on certain elements).

Solution: pp. 185 in the course book.

SOLUTION

Question 12 :

(6p) What is the **purpose** and **limitations** of using Field Experiments and Field Studies as a research strategy? Provide **examples**, i.e., methods for each of the two categories, and **clarify** if one use mostly qualitative or quantitative approaches (or both).

Solution:

Field Experiments: Natural settings.

Obtrusiveness: Researcher manipulates some variables/properties to observe an effect. Researcher does not control the setting.

Goal: Investigate/Evaluate/Compare techniques in concrete and realistic settings.

Limitations: Low statistical generalizability (claim analytical generalizability!) Fairly low precision of the measurement of the behavior (confounding factors).

Not the same as an experimental study (changes are made and results observed), but does not have control over the natural setting.

Correlation observed but not causation.

Examples: Evaluative case study, Quasi-experiment, and Action research.

Often qualitative and quantitative.

Field Studies: Natural settings.

Obtrusiveness: Researcher does not change/control parameters or variables.

Goal: Understand phenomena in concrete and realistic settings. Explore and generate new hypothesis.

Limitations: Low statistical generalizability (claim analytical generalizability!) Likely low precision of the measurement of the behavior.

Examples: Case study, Ethnographic study, and Observational study.

Often qualitative.

Question 13 :

(6p) Below follows an abstract from a research paper. Answer the questions,

- Which of the eight research strategies presented in the ABC framework does this paper likely fit? **Justify and argue!**
- Can you argue the main validity threats of the paper, based on the research strategy you picked?
 - It would be very good if you can **list threats in the four common categories** we usually work with in software engineering, i.e., internal, external, construct, and conclusion validity threats.

Software engineering is a human activity. Despite this, human aspects are under-represented in technical debt research, perhaps because they are challenging to evaluate.

This study's objective was to investigate the relationship between technical debt and affective states (feelings, emotions, and moods) from software practitioners. Forty participants ($N = 40$) from twelve companies took part in a mixed-methods approach, consisting of a repeated-measures ($r = 5$) experiment ($n = 200$), a survey, and semi-structured interviews.

The statistical analysis shows that different design smells (strong indicators of technical debt) negatively or positively impact affective states. From the qualitative data, it is clear that technical debt activates a substantial portion of the emotional spectrum and is psychologically taxing. Further, the practitioners' reactions to technical debt appear to fall in different levels of maturity.

We argue that human aspects in technical debt are important factors to consider, as they may result in, e.g., procrastination, apprehension, and burnout.

Solution: This is a mixed method study, i.e., a laboratory experiment, but where we also see strong qualitative parts, like from a field study.

Question 14 :

(11p) A common research method in software engineering that is used to complement other methods is survey research. Here follows a number of questions connected to survey research:

1. We often differ between reliability and validity concerning surveys,
 - (a) What is the difference between the reliability and validity in survey design? (2p)
 - (b) Name and describe at least two types of reliability in survey design. (3p)
 - (c) Name and describe at least two types of validity in survey design. (3p)
2. Even if you measure and estimate reliability and validity you still want to *evaluate* the survey instrument. Which are the two (2) common ways of evaluating a survey instrument? Explain their differences. (3p)

Solution:

1a. Reliability: A survey is reliable if we administer it many times and get roughly the same distribution of results each time. Validity: How well the survey instrument measures what it is supposed to measure?

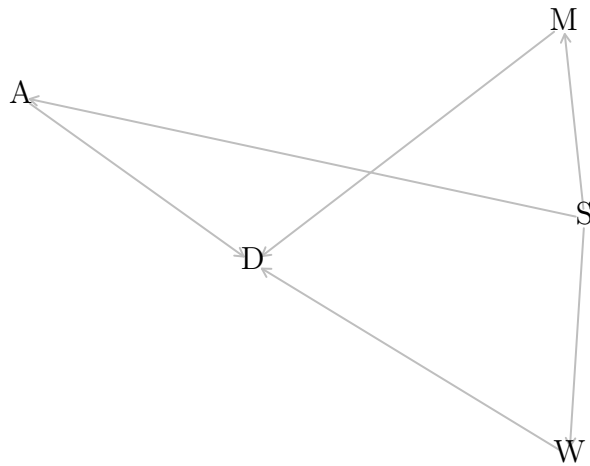
1b. e.g., Test-retest (intra-observer), alternate form reliability, internal consistency, inter-observer (Krippendorff's α).

1c. e.g., content validity, criterion validity, construct validity.

2. **Focus groups:** Assemble a group of people representing respondents, experts, beneficiaries of survey results. Ask them to complete the survey questionnaire. The group identifies missing or unnecessary questions, ambiguous questions and instructions. **Pilot studies:** Uses the same procedures as planned for the main study but with smaller sample. Targets not only problems with questions but also response rate and reliability assessment.

Question 15 :

(6p) See the DAG below. We want to estimate the total causal effect of W on D . Which variable(s) should we condition on and why?

**Solution:**

```
> dag <- dagitty("dag {  
  A -> D  
  A <- S -> W  
  S -> M -> D  
  S -> W -> D  
}")  
> adjustmentSets(dag, exposure = "W", outcome = "D")
```

and the answer is $\{ A, M \} \setminus \{ S \}$, i.e., either **A and M**, or only **S**. In short, check what elementary confounds we have in the DAG (DAGs can only contain four), follow the paths, decide what to condition on.